

As an Executive Director on the Central Risk Book (CRB) desk at Nomura, I evaluate candidates on their ability to move beyond the flawed Black-Scholes paradigm. In our equities business, we manage large portfolios where the "smile"—the empirical observation that implied volatility varies by strike—is not a theoretical anomaly; it is a P&L reality.

The Heston (1993) model is the industry standard for pricing these effects by treating variance as a stochastic process, rather than a constant parameter.

1. The Mathematical Deep Dive: Step-by-Painstaking-Step

Step 1: The Coupled Stochastic Differential Equations (SDEs)

The Heston model defines the joint dynamics of the asset price S_t and its variance V_t under the risk-neutral measure $\mathbb{Q}$:

1. Asset Price SDE:

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_t^S$$

This is essentially Black-Scholes, but with a stochastic variance term $\sqrt{V_t}$ replacing the constant volatility σ . 2.

Variance SDE (CIR Process):

$$dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^V$$

Here, V_t follows a **Cox-Ingersoll-Ross (CIR)** process. It is a mean-reverting square-root diffusion where κ is the speed of mean reversion, θ is the long-run variance, and ξ is the "volatility of volatility." 3. **Correlated Noise:**

$$dW_t^S dW_t^V = \rho dt$$

The correlation ρ between price shocks and volatility shocks is what creates the **volatility skew**. For equities, $\rho < 0$ is typical: as price drops, volatility spikes.

Step 2: The Feller Condition

Because we take $\sqrt{V_t}$ in the diffusion terms, if V_t reaches zero, the process could technically turn negative, which is mathematically undefined in this framework. To ensure $V_t > 0$ almost surely, the drift must be strong enough to push the process away from zero. This is the **Feller condition**:

$$2\kappa\theta > \xi^2$$

Step 3: Deriving the Pricing PDE via Feynman-Kac

Let $U(S, V, t)$ be the price of an option. By the Feynman-Kac theorem, the value of a derivative under a diffusion process must satisfy the Kolmogorov backward equation. Applying Itô's Lemma to $U(S, V, t)$ and enforcing the risk-neutral martingale condition, we arrive at the Heston PDE:

$$\frac{\partial U}{\partial t} + rS \frac{\partial U}{\partial S} + \kappa(\theta - V) \frac{\partial U}{\partial V} + \frac{1}{2} V S^2 \frac{\partial^2 U}{\partial S^2} + \rho \xi V S \frac{\partial^2 U}{\partial S \partial V} + \frac{1}{2} \xi^2 V \frac{\partial^2 U}{\partial V^2} - rU = 0$$

- $\frac{1}{2} V S^2 \frac{\partial^2 U}{\partial S^2}$: Gamma of the asset.

- $\rho\xi VS \frac{\partial^2 U}{\partial S \partial V}$: Cross-gamma term capturing the impact of correlation (skew).
- $\frac{1}{2}\xi^2 V \frac{\partial^2 U}{\partial V^2}$: Vol-gamma term capturing the curvature of the volatility surface.

Step 4: The Characteristic Function Approach

Solving the PDE directly is numerically intensive. Instead, Heston demonstrated that the log-price follows a structure that allows for a closed-form solution in the Fourier domain. The characteristic function is:

$$\phi(u) = \exp(C(T, u) + D(T, u)V_0 + iu \ln S_0)$$

The functions $C(T, u)$ and $D(T, u)$ are solutions to **Riccati ODEs**. They depend on the parameters $(\kappa, \theta, \xi, \rho, r)$ and are solved analytically. Once we have the characteristic function, we perform a **Fourier Inversion** (via the Gil-Pelaez formula) to recover the risk-neutral probability density, which is then integrated against the payoff to find the option price C .

2. Application: Why CRB Needs This

In the CRB, we don't just trade spot—we trade the **volatility surface**.

- **Volatility Smile/Skew**: Black-Scholes assumes a flat surface. Heston's ρ and ξ parameters allow us to fit market-observed skews.
- **Correlation Hedging**: Because we hedge residual risks, understanding how our delta changes as vol changes (Vanna) and as vol-of-vol changes (Volga) is mathematically captured by the second-order derivatives in the Heston PDE.

Production Implementation Note

When implementing Heston pricing, we use the **Lord-Fang integration approach** for the Fourier inversion to handle the numerical instability of the characteristic function at low maturities.

```
import numpy as np

def heston_char_func(u, T, r, kappa, theta, xi, rho, V0, S0):
    """
    Computes the Heston characteristic function (ruling out branch cuts).
    """
    d = np.sqrt((rho * xi * u * 1j - kappa)**2 + xi**2 * (u * 1j + u**2))
    g = (kappa - rho * xi * u * 1j - d) / (kappa - rho * xi * u * 1j + d)

    C = r * u * 1j * T + (kappa * theta / xi**2) * (
        (kappa - rho * xi * u * 1j - d) * T - 2 * np.log((1 - g * np.exp(-d * T))
        / (1 - g))
    )
    D = ((kappa - rho * xi * u * 1j - d) / xi**2) * ((1 - np.exp(-d * T)) / (1 - g
    * np.exp(-d * T)))

    return np.exp(C + D * V0 + u * 1j * np.log(S0))
```

3. Executive Director's Follow-up

Follow-up Question: "Heston assumes parameters are constant over time. However, CRB risk metrics clearly show that vol-of-vol (ξ) spikes during market crashes. If your model assumes ξ is constant, you are underpricing tail risk. How do you fix this without breaking the closed-form characteristic function?"

- **The Fix:** We cannot use a simple Heston model in isolation for high-stress CRB risk management. We shift to a **Piecewise Constant Heston** model, or we introduce a **jump-diffusion component** (Bates Model). The Bates model adds a Poisson jump process to the SDE:

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_t^S + J_t dN_t$$

- This captures the discontinuous "gap" risk observed in cash equities during limit-down scenarios, which Heston's pure diffusion cannot handle. The characteristic function simply becomes the product of the Heston characteristic function and the jump process characteristic function, preserving the analytical tractability of the Fourier inversion.